

In **SatQuery AI**, **Agentic AI** represents a shift from static, single-purpose remote-sensing models and generic VLM wrappers toward a **query-driven, autonomous remote-sensing intelligence system** 1-3.

Instead of relying on a single unadapted model or requiring non-expert users to navigate complex GIS workflows, an **agentic controller** automatically interprets natural-language queries, validates image inputs, orchestrates specialized AI tools, and synthesizes evidence-grounded responses 1, 4-6.

1. Paradigm Shift: From Generic VLM to Agentic System

Conventional multimodal AI systems typically follow a direct Image → VLM → Answer pipeline 3. However, unadapted VLMs frequently struggle with satellite imagery and suffer from tool hallucination or argument formatting errors 7, 8.

SatQuery AI replaces this with an end-to-end agentic workflow: **Query** → **Query Interpreter** → **Compatibility Validator** → **Specialist Router** → **Specialist Execution** → **Verifier Node** → **Evidence Fusion** → **Grounded Response** 3, 9, 10.

2. Core Architectural Principles

SatQuery AI's agentic framework is built around three primary design principles:

- **Typed Execution Graph ("Validate Before You Execute")**: Rather than using a fragile, free-form ReAct loop where an LLM guesses tools on the fly, SatQuery AI uses a deterministic, schema-validated task-graph 11-14.
- **Structural Precondition Checking**: A non-LLM **Compatibility Validator** evaluates machine-readable preconditions (such as sensor modality, CRS, spatial footprint overlap IoU, and temporal ordering) before any specialist model runs, structurally rejecting physically unsatisfiable queries 10, 12, 13, 15.
- **Auditable Verification & Evidence Fusion**: Specialist outputs are cross-checked by a **Verifier Node** performing deterministic geometric sanity checks (bounding box bounds, change-mask extents) and quantity cross-checks before generating the final response 10, 11, 16.

3. Key Agentic Operations

The agentic controller performs a structured 7-step execution sequence 10, 17:

1. **Query Interpretation**: A small instruct LLM translates natural-language queries into a typed TaskSpec JSON schema 10, 17, 18.
2. **Input Compatibility Validation**: Validates image count, file formats, sensor modalities, and georeferenced spatial overlap 10, 15, 17.
3. **Deterministic Specialist Routing**: Maps the validated TaskSpec to specialized tools (such as LoRA-adapted VQA, grounding, change detection, or optical-SAR fusion) via a fixed registry 10, 17, 19.
4. **Specialist Execution**: Executes task-specific LoRA adapters over a shared, frozen vision-language backbone 10, 20.
5. **Verification & Sanity Checking**: Verifies bounding boxes, checks cross-tool agreement, and compares stated numerical claims against deterministic pixel counts 10, 16.

6. **Confidence & Audit Trace Emission:** Assigns a High/Medium/Low confidence badge based on decoupled perception and reasoning signals and serializes a complete, auditable ExecutionTrace 10, 17, 21, 22.
7. **Response & Visual Evidence Composition:** Generates the final natural-language explanation alongside visual overlays (bounding boxes/change masks) citing the exact tool and sensor modality that supplied each claim 10, 17, 23, 24.

4. Why Agentic AI Matters in Remote Sensing

In Earth observation, queries often require joint reasoning across multiple sensor types (such as optical and SAR pairs) or multitemporal observations 4, 7, 25. Agentic AI allows non-expert users to extract actionable insights from complex satellite imagery through natural language while ensuring every output is verified, transparent, and fully traceable 4, 6, 26.

💡 *Would you like to review a step-by-step example of how the Compatibility Validator rejects an invalid query, or explore how the Verifier Node performs geometric sanity checks on change masks?*